Week 7 online class

Question 13 to 15, Worksheet 4

Question 13. Let $\left(a_k\right)$ and $\left(b_k\right)$ be two sequences such that

$$a_k = b_{k+1} - b_k \; \forall k \in \mathbb{N} \, .$$

Prove that $\sum_{k=1}^{\infty} a_k$ converges if and only if $\lim\left(b_k\right)$ exists,

and that $\sum_{k=1}^{\infty} a_k = \lim\left(b_k\right) - b_1 \, .$

Proof. Let $s_n = \sum_{k=1}^{n} a_k$. Then $s_n = \sum_{k=1}^{n} (b_{k+1} - b_k) = b_{n+1} - b_1$.

So, $\sum_{k=1}^{\infty} a_k$ converges $\Leftrightarrow$ $\left(s_n\right)$ converges $\Leftrightarrow$ $\lim\left(b_k\right)$ exists.

Suppose $\lim\left(b_k\right)$ exists, then $\sum_{k=1}^{\infty} a_k = \lim\left(s_n\right) = \lim\left(b_k\right) - b_1$.

Question 14. Let $a_k > 0 \ \forall k \in \mathbb{N}$.

Prove that if $\sum_{k=1}^{\infty} a_k^2$ converges, then $\sum_{k=1}^{\infty} \frac{a_k}{k}$ converges.

Proof. Let $a_r, b_r \geq 0$ for all $r \geq 1$.

First, we prove that if $\sum_{r=1}^{\infty} a_r^2$ and $\sum_{r=1}^{\infty} b_r^2$ converge, then $\sum_{r=1}^{\infty} a_r b_r$ is convergent.

From the fact

$$(a_r - b_r)^2 \geq 0 \Rightarrow \tfrac{1}{2}\left(a_r^2 + b_r^2\right) \geq a_r b_r \,,$$

and since $\tfrac{1}{2}\sum_{r=1}^{\infty}\left(a_r^2 + b_r^2\right)$ converges,

it follows by the comparison test that

$\sum_{r=1}^{\infty} a_r b_r$ converges.

Now let $b_r = \frac{1}{r}$.

Then $\sum_{r=1}^{\infty} b_r^2$ converges.

Given that $\sum_{r=1}^{\infty} a_r^2$ converges,

so is $\sum_{r=1}^{\infty} a_r b_r = \sum_{r=1}^{\infty} \frac{a_r}{r}$.

Question 15.

Let $\left(a_n\right)$ be a decreasing sequence.

(a) Prove that if $\lim\left(a_n\right) = 0$, then $a_n \geq 0 \ \forall n \in \mathbb{N}$.

(b) Hence, prove that if $\sum_{k=1}^{\infty} a_k$ is convergent, then $\lim\left(na_n\right) = 0$.

- Part (a) is proven in Week 5 lecture.

Theorem 4.1. Cauchy Criterion for Series

$$\sum_{k=1}^{\infty} a_k \text{ converges} \iff \forall \varepsilon > 0 \ \exists N \in \mathbb{N} \text{ such that } \left| \sum_{k=m+1}^{n} a_k \right| < \varepsilon \ \forall n > m > N$$

By Cauchy criterion, $\forall \varepsilon > 0 \ \exists N \in \mathbb{N}$ such that

$$\left| \sum_{k=m+1}^{n} a_k \right| < \tfrac{\varepsilon}{2} \ \forall n > m > N \, .$$

Since $\left(a_k\right)$ decreases and from (a), it follows that

$$a_m \geq a_{m+1} \geq \cdots \geq a_n \geq 0 \, .$$

Choose $n = 2m > m$ and so

$$ma_{2m} \leq \sum_{k=m+1}^{n} a_k = \left| \sum_{k=m+1}^{n} a_k \right| < \tfrac{\varepsilon}{2} \Rightarrow \left| na_n \right| = na_n = 2ma_{2m} < \varepsilon \, .$$